Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/17/2006
PI Name:	Chang, Hsueh-Chia Ph.D.		
Project Title:	Microcirculation Anomalies in Microgravity Blood Flow		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Hsueh-Chia.Chang.2@nd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	574 631 5697
Organization Name:	Notre Dame University		
PI Address 1:	165A Fitzpatrick Engr.		
PI Address 2:			
PI Web Page:			
City:	Notre Dame	**State:**	IN
Zip Code:	46556	**Congressional District:**	3
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/04/2002	**End Date:**	10/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Ansari, Rafat	**Contact Phone:**	216-433-5008
Contact Email:	Rafat.R.Ansari@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2701		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project is to study anomalies in micro-gravity blood flow in human circulation system and in blood diagnostic kits. We have solved an important blood suspension loading problem into small diagnostic kits and are now working on several chip-scale blood analytical tools. Included are blood coagulation detecion, blood glucose detection, leukemia, sickle cell and blood bacteria detection. The target is to produce simple kits that do not require elaborate equipment that is unavailable in space. To develop blood diagnostic kits that can be used in a confined spaceship. The diagnostic kits would be to detect pathogens and anomal physiological symptons duirng prolonged space flights. It can produce a robust and miniature bloood diagnostic kit for a variety of blood anomalies and diseases often associated with space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The same blood diagnostic kits could be used as point-of-care and portable diagnostic kits on earth.
Task Progress:	Research has been carried out to study anomalous blood rheology under microgravity conditions or under micro-flow conditions when blood cell sedimentation is insignificant. The research has led to the discovery of several new blood wetting phenomena that are caused by unhealthy blood cells. These wetting phenomena are unimportant when blood cell gravitational sedimentation is negligible but becomes very pronounced in micro-channel blood flow and in space. Such micro-channel blood flow occurs on earth in blood diagnostic kits. As such, our discoveries have led to better blood diagnostic designs and, in fact, new diagnostics for unhealthy blood cells. The anomalous blood wetting phenomena are driven by blood cell segregation under micro-flows of high shear rate. The segregation produces meniscus packing that can significantly retard the wetting advance. As the degree of blood cell segregation is a strong function of blood cell elasticity and blood cell size and shape, this segregation phenomenon has been used to detect septic, malarial and sickle blood cells, all of which involve deformed and inelastic infected cells. The study has also led to a new design of micro-needles for blood diagnostics that can handle sub-micro-liter blood samples. More tests are being done on the diagnostic application of this new phenomenon and the funding period has been extended by a year to October 31, 2006.
Bibliography Type:	Description: (Last Updated: 02/21/2014)
Articles in Peer-reviewed Journals	Zhou R, Gordon J, Palmer AF, Chang HC. "Role of erythrocyte deformability during capillary wetting." Biotechnol Bioeng. 2006 Feb 5;93(2):201-11. PMID: 16302256 , Feb-2006
Articles in Peer-reviewed Journals	Zhou R, Chang HC. "Capillary penetration failure of blood suspensions." J Colloid Interface Sci. 2005 Jul 15;287(2):647-56. PMID: 15925633 , Jul-2005